

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) now generate high-fidelity images, videos, and other structured outputs, but perceptual realism does not ensure adherence to factual regularities in the data: a sample may look plausible while containing an inadmissible duplicate object or an anatomically impossible hand. A central obstacle to studying such failures is the lack of controlled, reproducible protocols that convert hallucination from a qualitative judgment into a measurable event. This paper develops a quantitative framework for a concrete and consequential subclass of these errors, which we term **counting hallucinations**: generated samples whose numbers of object instances or structured components fall outside the count constraints encoded in the reference data. We introduce **CountHalluSet**, a dataset suite with explicit counting criteria across ToyShape, SimObject, and RealHand, and instantiate a standardized evaluation protocol that distinguishes counting-valid samples, counting hallucinations, and non-counting visual failures. Using this framework, we systematically examine how solver family, ODE solver order, sampling budget, and initial noise configuration affect counting hallucination rates in DPMs. We further show, through correlation analyses with perceptual metrics including Fréchet Inception Distance (FID), that widely used image-quality measures do not consistently characterize count-level factual errors, even though they more reliably track non-counting failures.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, have become a dominant family of generative models, producing high-quality samples for text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Their impressive perceptual fidelity, however, should not be conflated with faithful reproduction of the factual regularities present in the training distribution. Diffusion models can synthesize outputs that are locally plausible yet globally inconsistent with real-world or dataset-level structure, including objects with an inadmissible number of spatially intricate components, such as extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

A growing literature has documented failure modes in text-guided diffusion systems. For example, Borji (2023) organized qualitative shortcomings of text-to-image models, Liu et al. (2024) introduced adversarial search for exposing undesirable generation behaviors, and Wu et al. (2023) developed an iterative procedure for discovering and mitigating spurious correlations. These studies identify important sources of model unreliability, but they do not provide a controlled quantitative account of hallucinations that persist even when prompt-output alignment is not the principal object of evaluation.

The notion of *hallucination* in diffusion models has therefore become an important research target (Aithal et al., 2024). Unlike prompt-alignment failures, which can be approximated through condition-output metrics such as CLIP score (Hessel et al., 2021) and VQA score (Huang et al., 2025; Zarei et al., 2024), and unlike conspicuous degradations such as severe blur or artifacts, hallucinated samples (*hallucinations*) may remain visually plausible while violating factual constraints implicit in the training data. Such errors include incorrect object counts, plausible-looking but physically impossible geometries, and other breaches of physical regularity. Because the boundary between

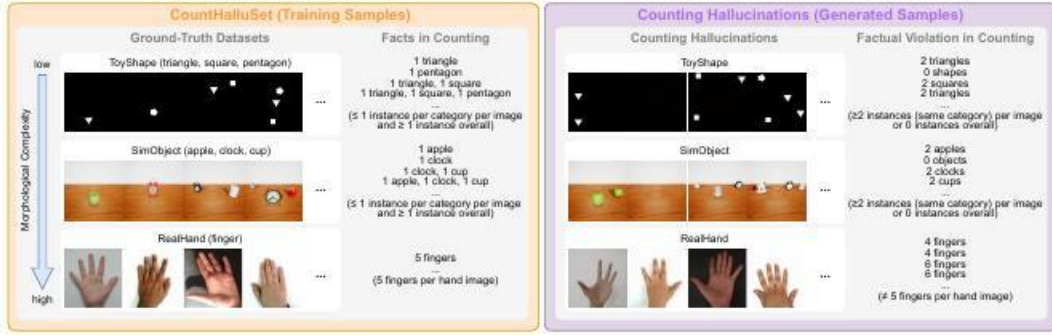


Figure 1: Representative training samples from the proposed **CountHalluSet** (left) and model-generated **counting hallucinations** evaluated under the corresponding dataset criteria (right). Each dataset specifies an explicit count constraint: ToyShape and SimObject permit at most one instance per category and require at least one object in each image, whereas RealHand requires exactly five fingers. A counting hallucination is therefore a generated image that violates the relevant dataset rule, such as two apples or an otherwise normal empty-background image in SimObject, although such configurations are absent from the training data.

a valid sample and a hallucination can be fine-grained, current AI systems often fail to identify it reliably; even state-of-the-art large vision-language models such as GPT-4o can struggle to count fingers in a clear hand image. A standardized measurement protocol is therefore necessary: it converts an otherwise qualitative failure mode into reproducible evidence and supports the design of generative models whose outputs preserve factual consistency, a prerequisite for their broader role as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among the many factual attributes that can be hallucinated—including counts, colors, spatial relations, geometry, and physical realism—object count is especially suitable for rigorous measurement because the admissible set of values can be specified before generation and evaluated after generation. We therefore focus on **counting hallucinations**. *In this setting, the number of objects in training data is explicitly predetermined. Consequently, a generated image is deemed hallucinated whenever its object counts deviate from these predefined values.* This definition is intentionally dataset-relative: a hand image with six fingers is a counting hallucination for RealHand because that configuration is excluded by the dataset construction, as illustrated in the last row of Fig. 1. To examine this failure mode across progressively richer visual structure, we construct CountHalluSet, a three-part suite comprising ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 shows representative valid samples and their corresponding counting hallucinations.

Our evaluation protocol operationalizes counting hallucination by combining explicit dataset-level count criteria with dataset-specific counting models and, for RealHand, a counting-readiness filter that separates count-evaluable images from severe visual failures. Within this framework, we quantify how denoising conditions in diffusion models—sampling steps, initial noise, ODE solver order, and solver type—alter counting hallucination rates. We also test whether standard perceptual metrics, including Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021), reflect these factual errors. The paper makes three central contributions:

- We define counting hallucination as a measurable, dataset-relative factual-consistency failure and instantiate it through CountHalluSet, whose three datasets impose explicit count constraints across synthetic shapes, rendered objects, and real hand images.
- We provide a controlled empirical characterization of how sampling steps, initial noise, solver order, and solver family affect counting hallucinations, showing that improvements observed on simpler synthetic datasets do not reliably transfer to human-hand generation.
- We show that counting hallucination rate is weakly and inconsistently associated with perceptual metrics, whereas non-counting failure rate is strongly associated with those metrics; this separation motivates a joint-diffusion training paradigm that reduces failure rates on RealHand.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

The extensive literature on hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023) motivates analogous questions for diffusion models, where factual errors are expressed through visual structure rather than text alone. For unconditional diffusion models, Aithal et al. (2024) advanced a qualitative hypothesis that hallucinated samples may arise through interpolation between distinct data modes. That perspective identifies a plausible mechanism, but the field still lacks a systematic protocol for measuring when, how often, and under which sampling conditions diffusion-generated images violate factual constraints.

This paper addresses that measurement gap by making a visually subtle hallucination class amenable to direct quantification. By isolating count violations and evaluating them under controlled denoising configurations, the study connects hallucination severity to concrete sampling choices such as step count and ODE solver order, thereby moving the analysis from anecdotal failure examples toward reproducible empirical evidence.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

VQA score (Huang et al., 2025; Zarei et al., 2024) and CLIP score (Hessel et al., 2021) are widely used to evaluate text-guided image generation, especially when the relevant question is whether an image matches its textual condition. Their reliance on language-based models, however, makes them an incomplete basis for hallucination assessment. Pretrained language models have a statistical lower bound on hallucinating certain fact types, which limits the accuracy attainable by detectors that depend exclusively on such models (Kalai & Vempala, 2024); moreover, human annotators continue to outperform LLMs in classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023).

Image generation is also commonly evaluated with quantitative perceptual metrics, including FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021). These metrics summarize distributional or image-quality properties through feature embeddings, often extracted from pretrained networks such as Inception-v3 (Szegedy et al., 2016). Whether such summaries capture factual count violations is an empirical question rather than a consequence of their design. Our experiments therefore measure the relationship between counting hallucinations and standard image-generation metrics directly, rather than assuming that better perceptual quality entails fewer hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (a.k.a, *diffusion process*) that progressively corrupts the data with Gaussian noise over T discrete timesteps. At timestep t , the conditional marginal distribution of the noisy variable $\mathbf{x}_t$ given the original sample $\mathbf{x}_0$ is

$$q_{0:t}(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t \mid \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_i\}_{i=1}^T$ is a predefined time-dependent noise schedule, commonly linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021). The coefficient $\bar{\alpha}_t$ controls the signal-to-noise ratio: $\bar{\alpha}_0 = 1$ corresponds to uncorrupted data, whereas $\bar{\alpha}_T \approx 0$ corresponds to nearly pure noise. Equivalently, Eq. (1) can be written as

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + (1 - \bar{\alpha}_t)\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

For sufficiently large T , the terminal marginal $q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ approaches the standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. Widely used DPMs typically set T to 1000 (Ho et al., 2020; Nichol & Dhariwal, 2021).

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (a.k.a., *denoising process*) learns to recover data samples by iteratively transforming the terminal noisy variable $\mathbf{x}_T$ back toward the data distribution. DPMs parameterize this process with a neural noise predictor $\epsilon_\theta(\mathbf{x}_t, t)$ that estimates the injected noise ϵ from the noisy input $\mathbf{x}_t$ at timestep t . The parameters θ are optimized with the standard denoising objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function.

Ancestral Sampling. Given terminal noisy data $\mathbf{x}_T$, samples from $q_0(\mathbf{x}_0)$ are generated by applying the learned reverse transition over the same discrete time horizon; this procedure is commonly called *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} \mid \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the unknown *score function* (Song et al., 2022).

In practice, DPMs replace the scaled score $-\sqrt{(1-\bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ with the noise prediction model trained by Eq. (3). Under this parameterization, the sampling update is

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. Ancestral sampling in Eq. (4) includes a stochastic term (Kloeden et al., 1992). The same marginal distributions can be represented by a deterministic probability-flow process, formulated as the following ordinary differential equation (ODE) (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory satisfying this ODE under the true score function $h(\mathbf{x}(\tau), \tau)$, initialized from ideal diffused data $\mathbf{x}(T)$ drawn from $q_T(\mathbf{x}(T))$. Define discrete times $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, with $\tau_N = 0$. Applying the variation-of-constants formula to one exact solver step yields

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state-transition operator for the linear subsystem $\frac{dz}{d\tau} = f(\tau)z$. The step size $\Delta t > 0$ determines the number of sampling steps, $T/\Delta t$, and larger steps enable ODE-based solvers to accelerate generation relative to ancestral sampling. Because $\mathcal{G}(t_2, t_1)$ has a closed-form expression, accelerated ODE samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) concentrate their numerical approximation error—as governed by solver order and step count—in the nonlinear integral involving h . Appendix A.1 to A.3 provides the accompanying theoretical analysis.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section formalizes the measurement framework used throughout the paper. We first define dataset-level counting criteria and the corresponding notion of a counting hallucination, then instantiate the definition in CountHalluSet. For each dataset, we specify the admissible counts, the violations recorded as counting hallucinations, and the standardized procedure used to quantify those violations in generated samples.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We define counting criteria as explicit constraints on the object counts that may appear in a reference dataset, and we define counting hallucinations as generated samples that violate those constraints while remaining visually suitable for count evaluation.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the count of category c in $\mathbf{x}$, written $N_c(\mathbf{x})$, is constrained to lie in a predefined admissible set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), so that $N_c(\mathbf{x}) \in \mathcal{S}_c$. We additionally require every valid reference sample to contain at least one object, $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C denotes the category set under evaluation.

Counting hallucinations as deviations from the counting criteria. For a generated sample $\hat{\mathbf{x}}$, we say that $\hat{\mathbf{x}}$ exhibits a *counting hallucination* relative to $\mathcal{D}_{\text{ref}}$ when the following two conditions are simultaneously satisfied:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ must be visually interpretable, with the background and any present objects sufficiently discernible for count evaluation. This requirement separates counting hallucinations from severe non-counting failures such as strong degradation or unrecognizable structure. We encode it with a binary *counting-ready indicator*, $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, where 1 denotes that the image is suitable for counting and 0 denotes that it is not. In practice, this indicator may be obtained through human annotation, a classifier calibrated to human judgments, or other automated criteria.
2. **Violation of counting facts.** A counting-fact violation occurs when some category c satisfies $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or when the generated image contains no objects (i.e., $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$) despite the reference criterion requiring at least one object. We operationalize this condition by applying a counting model to generated images and using its predicted counts as evidence of violations.

A counting hallucination (CH) is therefore recorded exactly when the sample is counting-ready ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and its predicted counts depart from the counting criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

CountHalluSet comprises three datasets that pose the same factual question—whether generated samples preserve admissible counts—under progressively richer visual structure: ToyShape, SimObject, and RealHand. Representative valid samples and corresponding hallucinated generations are provided in Fig. A1 and A2, with additional dataset statistics reported in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. ToyShape contains 30,000 images composed from three geometric categories: *triangle*, *square*, and *pentagon*. Each shape is white, occupies 120 pixels in area, and is placed without overlap. The counting criterion is deliberately unambiguous: each image contains at most one instance of each category and at least one total shape, as shown in the top-left of Fig. 1. Formally, $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under these criteria, any sample generated by a model trained on ToyShape that contains two or more instances of the same shape category is a counting hallucination, such as the two-pentagon example in the top-right region of Fig. 1. Empty images, consisting only of the black background and no shapes, also violate the dataset criterion and are recorded as hallucinations. Additional examples appear in Appendix A.5.

Counting model used in ToyShape. Because ToyShape rarely produces severe visual failures, generated images are evaluated directly with a counting model and no counting-ready indicator is applied. Specifically, we fine-tune a ResNet-50 (He et al., 2016) on more than 400k ToyShape images

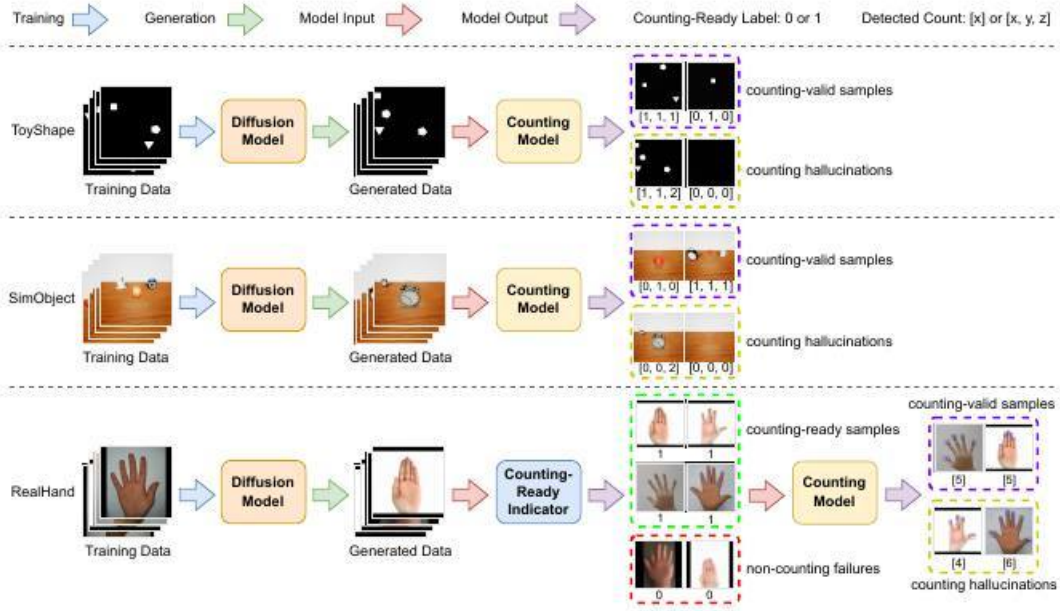


Figure 2: Evaluation procedure for CountHalluSet. ToyShape and SimObject generations are passed directly to dataset-specific counting models because severe non-counting failures are rare in these controlled settings. RealHand generations first pass through a counting-ready indicator, implemented as a binary classifier aligned with human judgments, to separate count-evaluable hand images from non-counting failures such as severely deformed fingers. Count labels are ToyShape: *triangle*, *square*, *pentagon*; SimObject: *mug*, *apple*, *clock*; RealHand: *finger*.

in which each category appears 0–3 times per sample (i.e., $S_c = \{0, 1, 2, 3\}$), so that the model observes both counting-valid and count-violating configurations during training. This model achieves over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. For each diffusion model trained on ToyShape, we generate 30,000 samples, matching the training-set size. The counting model predicts the number of objects in each generated image, and any image whose predicted counts violate the ToyShape criteria is recorded as a counting hallucination, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. SimObject contains 30,000 rendered images of everyday objects from three categories: *mug*, *apple*, and *clock*, with 10 intra-class variants per category. Objects are randomly placed on a wooden tabletop under fixed illumination and with minimal mutual occlusion, as illustrated in the middle-left panel of Fig. 1. The images are generated in Unreal Engine 5 (Epic Games, 2024), providing photorealistic renderings of object geometry, material properties, lighting, and reflections. The counting criteria match the ToyShape structure (see Sec. 4.2.1): $C = \{mug, apple, clock\}$, and for each class $c \in C$, $N_c(\mathbf{x}) \in S_c = \{0, 1\}$ under the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Because SimObject uses the same admissible-count structure as ToyShape, its counting hallucination definition, counting-model construction, and quantification protocol follow the ToyShape procedure, as shown in Fig. 2.

4.2.3 REALHAND

Dataset description. RealHand evaluates counting hallucinations in real hand imagery. It contains 5,050 human hand images collected from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². After aggregating the raw images, we remove samples with blur or visible artifacts. Each retained image depicts a five-finger human hand, including both dorsal (back) and palmar (front) views under

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

diverse lighting conditions and backgrounds such as white surfaces, blackboards, and walls, as shown in the lower-left panel of Fig. 1. The counting criterion is exact: $C = \{finger\}$ with $N_c(\mathbf{x}) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(\mathbf{x}) = 5$.

Counting hallucinations and non-counting failures. Hand generation introduces failure modes beyond count errors, including blur, severe finger deformation, and other visual artifacts. We therefore distinguish counting hallucinations, such as extra or missing fingers in an otherwise countable image, from non-counting failures that are unsuitable for reliable count evaluation. This separation is implemented with a counting-ready indicator, and examples of both categories are provided in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To separate count-evaluable hands from non-counting failures, we finetune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images as a counting-ready indicator aligned with human judgments. For finger counting, we finetune a YOLO-12 (Tian et al., 2025) detector on 2k fingertip-annotated generated images. On 500 randomly sampled images evaluated by three human annotators, the indicator and counting model achieve over 96% and 99% accuracy, respectively. Appendix A.6 shows representative YOLO detection results.

Quantification of counting hallucinations. Unlike ToyShape and SimObject, where severe non-counting failures are rare and all generated images are evaluated directly, RealHand requires an explicit filtering stage. We first apply the counting-ready indicator to select images with sufficient visual quality and then apply the counting model to estimate the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. For ToyShape, we train a DDPM (Ho et al., 2020) from scratch; for SimObject and RealHand, we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) initialized from models pretrained on CelebA-HQ (Karras et al., 2018). All diffusion models use $T = 1,000$ timesteps with a standard linear noise schedule. Images are resized to 128×128 for DDPM training and to 256×256 for LDM training. Optimization uses Adam (Kingma, 2015) with learning rate 0.0001 and effective batch size 256; training runs for 150k, 300k, and 80k steps on ToyShape, SimObject, and RealHand, respectively.

Inference of Diffusion Models. Ancestral sampling (DDPM) uses 1,000 sampling steps. For accelerated ODE-based sampling, we evaluate DPM-Solver-1, equivalent to DDIM (Song et al., 2021), as the first-order solver and DPM-Solver-2 (Lu et al., 2022) as the second-order solver. Each ODE solver is tested at 25, 50, and 100 denoising steps. For RealHand, the YOLO counting model uses confidence threshold 0.3 and IoU threshold 0.1. Unless otherwise specified, each evaluation generates the same number of samples as the corresponding training set, and reported results are averaged over three random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) across solver types, ODE solver orders, sampling-step counts, and initial-noise settings for ToyShape, SimObject, and RealHand. These controlled comparisons make five empirical patterns explicit.

(1) Increasing sampling steps reduces counting hallucinations in synthetic datasets but not in RealHand. As shown in Table 1, under ODE sampling with 25, 50, and 100 steps, CHR generally decreases on ToyShape and SimObject as the number of steps increases, whereas CHR tends to increase on RealHand, except for the 100-step DPM-Solver-2 condition. This contrast indicates that finer solver discretization benefits the simpler, more structured synthetic distributions, but does not transfer straightforwardly to morphologically complex hand imagery, where additional denoising steps may coincide with amplified count-level inconsistencies.

Table 1: Counting hallucination rates (CHR; counting hallucinations divided by total generated samples) under solver configurations and initial-noise settings across ToyShape, SimObject, and RealHand. For RealHand, we additionally report non-counting failure rate (NCFR), which measures failure cases captured by the counting-ready indicator. The total failure rate (TFR) is defined as CHR plus NCFR. *Diffused* and *Normal* denote ground-truth initial noise and standard Gaussian noise, respectively.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

(2) **Higher-order ODE solvers reduce non-counting failures while increasing counting hallucinations in RealHand under standard inference settings.** As shown in Table 1, on RealHand at standard sampling budgets such as 25 and 50 steps, DPM-Solver-2 reduces NCFR relative to DPM-Solver-1 but produces higher CHR. The same qualitative trade-off appears when increasing sampling steps for both solvers. Thus, sampling conditions that improve overall visual adequacy, as reflected by lower non-counting failure rates, do not necessarily preserve the factual correctness of finger counts.

(3) **DDPM achieves the lowest failure rates across the evaluated configurations.** Across the reported comparisons, ancestral sampling (DDPM) yields the lowest CHR, NCFR, and TFR. Within this experimental setting, DDPM is therefore the strongest evaluated sampling strategy for reducing both counting hallucinations and non-counting failures.

(4) **A better initial noise configuration can reduce both counting hallucinations and non-counting failures.** As shown in Table 1, using “diffused” noise (i.e., ground-truth initial noise) tends to yield lower CHR, NCFR, and TFR than using “normal” noise (i.e., standard Gaussian noise). This pattern is consistent with the interpretation that diffused noise better matches the initial-noise distribution encountered during training, so a more informed initialization can mitigate both count-specific hallucinations and broader non-counting failures.

(5) **Greater morphological complexity in countable objects increases the likelihood of counting hallucinations.** Comparing CHR across ToyShape, SimObject, and RealHand shows that the evaluated diffusion models exhibit higher CHR as the countable structure becomes morphologically richer. The result indicates that fine-grained parts and complex spatial dependencies make count preservation substantially harder, producing more frequent additions or omissions in generated samples.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Because counting hallucination is a factual-consistency failure, a central empirical question is whether stronger perceptual scores—for example, lower FID values that suggest a smaller distributional gap between generated and reference images—also imply fewer count violations. Our results show that they do **not necessarily**.

To test this relationship directly, we estimate correlations between CHR and perceptual metrics and, separately, between NCFR and the same metrics across the evaluated sampler and step configurations. The analysis uses two widely adopted perceptual measures: FID (Heusel et al., 2017), a distribution-level metric, and MUSIQ (Ke et al., 2021), an instance-level image-quality metric. We report Pearson correlation (Lee Rodgers & Nicewander, 1988) for linear dependence, Spearman rank correlation (Spearman, 1904) for monotonic dependence, and Distance Correlation (Székely et al., 2007) for general nonlinear dependence. Because these configurations are structured rather than a large set of independent experimental conditions, the correlations should be read as diagnostic summaries of observed trends rather than as exhaustive statistical claims. Together, these complementary statistics make the distinction between factual count errors and perceptual failures empirically explicit.

Table 2: Correlations between counting hallucination rate (CHR), non-counting failure rate (NCFR), total failure rate (TFR), and perceptual metrics, including FID and MUSIQ. PTI-Hand results are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the size of the RealHand dataset. FID for PTI-Hand is computed between RealHand images and the model-generated samples. Here, p denotes the p-value associated with the reported correlation r or ρ . $\dagger$ indicates that DDPM results are integrated with those from various ODE solvers across sampling conditions.

Metric	CHR vs. FID				NCFR vs. FID				CHR vs. MUSIQ				NCFR vs. MUSIQ			
	SimObject	RealHand	PTI-Hand		RealHand	PTI-Hand			RealHand	PTI-Hand			RealHand	PTI-Hand		
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01
Pearson † (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01
Spearman † (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—
Distance Correlation † (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—

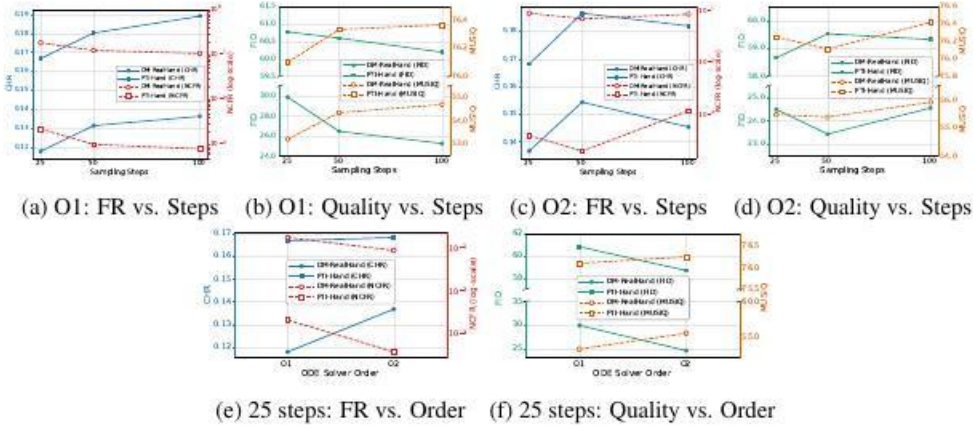


Figure 3: Trend comparison between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). FR metrics include CHR and NCFR; quality metrics include FID and MUSIQ. Across variations in sampling steps and solver order, the two settings exhibit closely aligned empirical trends.

✎ **The correlations between CHR and perceptual metrics are dataset-dependent and solver-dependent rather than intrinsic.** As shown in the first and third rows of Table 2, the Pearson correlations between CHR and FID vary substantially across datasets (i.e., positive for SimObject, negative for RealHand, and positive again for PTI-Hand), while the corresponding Spearman correlations shift from positive to negative to negative. Distance correlations also decrease from 0.82 (SimObject) to 0.74 (RealHand) to 0.47 (PTI-Hand), indicating that the association between CHR and FID is not stable across generative settings. Moreover, integrating DDPM results with those from ODE solvers further attenuates these correlations.

✎ **The correlations between NCFR and perceptual metrics are consistently strong and positive.** In contrast to the weak and unstable CHR–metric associations, NCFR exhibits strong positive correlations with perceptual metrics across all reported settings. For NCFR versus FID, all correlation coefficients exceed 0.88 and the corresponding p-values are very small, indicating that the automated non-counting failure measure aligns much more closely with perceptual degradation than CHR does.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

This section evaluates whether the RealHand protocol transfers to a large-scale pretrained text-to-image model. Specifically, we assess Stable-Diffusion-3.5-Medium (Esser et al., 2024) using the counting-ready indicator and counting model introduced in Sec. 4.2.3. More details are provided in Appendix A.8. As shown in Figure 3, the effects of solver type and sampling steps on counting hallucinations in the pretrained model closely align with the trends observed in the controlled

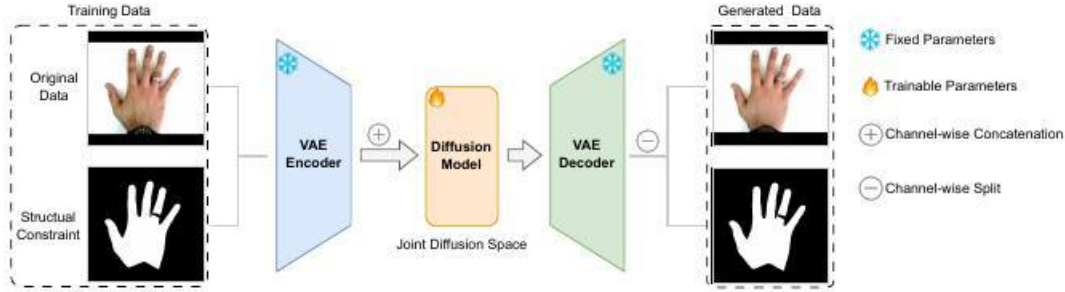


Figure 4: Diagram of the proposed joint-diffusion model (JDM). The model performs channel-wise concatenation of the latent representations of hand RGB images and their corresponding segmentation masks. This joint learning paradigm supplies explicit structural information during denoising and, in the RealHand experiments, reduces CHR and NCFR.

RealHand experiments. This consistency supports the applicability of the evaluation protocol beyond models trained only within CountHalluSet, while remaining within the hand-generation setting studied here. Correlation results in Table 2 further support this conclusion: CHR again shows weak or unstable associations with perceptual metrics, whereas NCFR is strongly associated with those metrics. Together, these results show that the proposed evaluation protocol transfers to pretrained text-to-image hand generation and that the empirical separation between count-level factuality and perceptual quality is not confined to the controlled RealHand model.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The lower counting hallucination rates observed for simpler countable objects (see Observation 5 in Sec. 5.2) suggest that diffusion models more reliably capture counting-based factuality when the underlying visual structure has lower morphological complexity, such as in mask-like representations. This observation motivates a mitigation strategy that is deliberately evaluated as an initial intervention rather than a fully controlled ablation: denoise general visual representations together with explicit structural constraints, such as object masks that encode basic topology and counts, within a shared latent space. We refer to models that jointly learn general visual representations and structural constraints during denoising as **joint-diffusion models (JDM)**.

Specifically, we perform a channel-wise concatenation of the latent representation of hand mask with that of the corresponding RGB image, as shown in Fig. 4. Then we finetune the UNet pretrained on CelebA-HQ (Karras et al., 2018) (following the RealHand configuration described in Sec. 5.1) using the RealHand images and their associated masks, extracted using SAM-2 (Ravi et al., 2024). To accommodate both modalities, the input and output channels of the denoising UNet are expanded to match the combined latent dimensionality of the image and mask latents (i.e., $3+3=6$ channels).

As shown in Table 3, JDM achieves lower failure rates than the vanilla LDM, including both CHR and NCFR, across the reported sampling conditions. This result indicates that auxiliary structural supervision can improve generation of morphologically intricate objects that are prone to counting ambiguities. At the same time, JDM yields higher FID. This degradation is consistent with the increased denoising burden introduced by jointly modeling RGB images and masks under a small-scale training regime, although the experiment does not isolate mask supervision from all associated changes in training budget and channel dimensionality. We also observe reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning settings. More details, results and discussion can be found in Appendix A.9.

6 CONCLUSION

This work establishes a systematic framework for quantifying a concrete factual-consistency failure in diffusion models: counting hallucination. By introducing **CountHalluSet** and an evaluation protocol that separates counting-valid samples, counting hallucinations, and non-counting failures, the paper turns count-level hallucination into a controlled empirical object. Across ToyShape, SimObject,

Table 3: Comparison of failure rates between LDM and the proposed JDM on RealHand across ODE solvers and sampling steps. Lower values indicate better performance. Although JDM jointly generates hand RGB images and segmentation masks, FID is computed only on RGB outputs to maintain comparability with the LDM baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

and RealHand, the experiments show that numerical sampling choices can reduce count violations in simpler settings but do not reliably mitigate them for human-hand generation. The correlation analyses further demonstrate that widely used perceptual metrics, including FID and MUSIQ, do not consistently reflect the severity of counting hallucinations, even when they align strongly with non-counting failures. Experiments with pretrained text-to-image hand generation corroborate the transferability of the protocol within this domain, and the joint-diffusion results indicate that explicit structural supervision is a promising direction for reducing count-level factual errors.

REFERENCES

- Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J Ballard, Joshua Bambrick, et al. Accurate structure prediction of biomolecular interactions with alphafold 3. *Nature*, pp. 493—500, 2024.
- Mahmoud Afifi. 11k hands: Gender recognition and biometric identification using a large dataset of hand images. *Multimedia Tools and Applications*, 78:20835–20854, 2019.
- Sumukh K Aithal, Pratyush Maini, Zachary C Lipton, and J Zico Kolter. Understanding hallucinations in diffusion models through mode interpolation. *arXiv preprint arXiv:2406.09358*, 2024.
- Ali Borji. Qualitative failures of image generation models and their application in detecting deepfakes. *Image and Vision Computing*, 137:104771, 2023.
- Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, et al. Genie: Generative interactive environments. In *ICML*, 2024.
- Thomas M Cover and Joy A Thomas. *Elements of Information Theory*. John Wiley & Sons, 2012.
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *NeurIPS*, 34:8780–8794, 2021.
- Jingtao Ding, Yunke Zhang, Yu Shang, Yuheng Zhang, Zefang Zong, Jie Feng, Yuan Yuan, Hongyuan Su, Nian Li, Nicholas Sukiennik, et al. Understanding world or predicting future? a comprehensive survey of world models. *ACM Computing Surveys*, 2024.
- Epic Games. Unreal engine, 2024. URL <https://www.unrealengine.com>.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *ICML*, pp. 12606–12633, 2024.
- David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2(3), 2018.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pp. 770–778, 2016.
- Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *EMNLP*, pp. 7514–7528, 2021.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *NeurIPS*, 30, 2017.

- Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, pp. 1–14, 2021.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 33: 6840–6851, 2020.
- Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. Video diffusion models. *NeurIPS*, 35:8633–8646, 2022.
- Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- Kaiyi Huang, Chengqi Duan, Kaiyue Sun, Enze Xie, Zhenguo Li, and Xihui Liu. T2i-compbench++: An enhanced and comprehensive benchmark for compositional text-to-image generation. *IEEE TPAMI*, 2025.
- Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*, 2023.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM computing surveys*, 55(12):1–38, 2023.
- John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873):583–589, 2021.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Adam Tauman Kalai and Santosh S Vempala. Calibrated language models must hallucinate. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, pp. 160–171, 2024.
- Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. Progressive growing of gans for improved quality, stability, and variation. In *ICLR*, 2018.
- Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *ICCV*, pp. 5148–5157, 2021.
- Diederik P Kingma. Adam: A method for stochastic optimization. In *ICLR*, pp. 1–15, 2015.
- Peter E Kloeden, Eckhard Platen, Peter E Kloeden, and Eckhard Platen. *Stochastic differential equations*. Springer, 1992.
- Zhifeng Kong, Wei Ping, Jiaji Huang, Kexin Zhao, and Bryan Catanzaro. Diffwave: A versatile diffusion model for audio synthesis. In *ICLR*, pp. 1–17, 2021.
- Joseph Lee Rodgers and W Alan Nicewander. Thirteen ways to look at the correlation coefficient. *The American Statistician*, 42(1):59–66, 1988.
- Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. Halueval: A large-scale hallucination evaluation benchmark for large language models. In *EMNLP*, pp. 6449–6464, 2023.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *ACL*, pp. 3214–3252, 2022.
- Luping Liu, Yi Ren, Zhijie Lin, and Zhou Zhao. Pseudo numerical methods for diffusion models on manifolds. In *ICLR*, pp. 1–24, 2022.
- Qihao Liu, Adam Kortylewski, Yutong Bai, Song Bai, and Alan Yuille. Discovering failure modes of text-guided diffusion models via adversarial search. In *ICLR*, pp. 1–25, 2024.

- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps. *NeurIPS*, 35:5775–5787, 2022.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver++: Fast solver for guided sampling of diffusion probabilistic models. In *ICLR*, pp. 1–19, 2023.
- Shitong Luo and Wei Hu. Diffusion probabilistic models for 3d point cloud generation. In *CVPR*, pp. 2837–2845, 2021.
- Supreeth Narasimhaswamy, Uttaran Bhattacharya, Xiang Chen, Ishita Dasgupta, Saayan Mitra, and Minh Hoai. Handdiffuser: Text-to-image generation with realistic hand appearances. In *CVPR*, pp. 2468–2479, 2024.
- Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *ICML*, pp. 8162–8171, 2021.
- Michael A Nielsen. *Neural networks and deep learning*, volume 25. Determination press San Francisco, CA, USA, 2015.
- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, pp. 10684–10695, 2022.
- Mehdi SM Sajjadi, Olivier Bachem, Mario Lucic, Olivier Bousquet, and Sylvain Gelly. Assessing generative models via precision and recall. *NeurIPS*, 31, 2018.
- Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training gans. *NeurIPS*, 29, 2016.
- Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *ICML*, pp. 2256–2265, 2015.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *ICLR*, pp. 1–22, 2021.
- Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *ICLR*, pp. 1–36, 2022.
- C Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.
- Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *CVPR*, pp. 2818–2826, 2016.
- Gábor J Székely, Maria L Rizzo, and Nail K Bakirov. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, pp. 2769–2794, 2007.
- Yunjie Tian, Qixiang Ye, and David Doermann. Yolov12: Attention-centric real-time object detectors. *arXiv preprint arXiv:2502.12524*, 2025.
- Zhengzhong Tu, Hossein Talebi, Han Zhang, Feng Yang, Peyman Milanfar, Alan Bovik, and Yinxiao Li. Maxvit: Multi-axis vision transformer. In *ECCV*, pp. 459–479, 2022.
- Arash Vahdat, Francis Williams, Zan Gojcic, Or Litany, Sanja Fidler, Karsten Kreis, et al. Lion: Latent point diffusion models for 3d shape generation. *NeurIPS*, 35:10021–10039, 2022.
- Shirley Wu, Mert Yuksekgonul, Linjun Zhang, and James Zou. Discover and cure: Concept-aware mitigation of spurious correlation. In *ICML*, pp. 37765–37786, 2023.

-
- Dmitry Yarotsky. Universal approximations of invariant maps by neural networks. *Constructive Approximation*, 55(1):407–474, 2022.
- Arman Zarei, Keivan Rezaei, Samyadeep Basu, Mehrdad Saberi, Mazda Moayeri, Priyatham Kattakinda, and Soheil Feizi. Understanding and mitigating compositional issues in text-to-image generative models. *arXiv preprint arXiv:2406.07844*, 2024.
- Qinsheng Zhang and Yongxin Chen. Fast sampling of diffusion models with exponential integrator. In *ICLR*, pp. 1–33, 2023.

A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the true reverse process $p_\theta(\mathbf{x}_{T:0})$, their KL divergence is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

For both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$, we can decompose them as follows, leveraging the Markov property:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

By applying the chain rule to decompose the KL divergence, we have:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

Then, according to the Data Processing Inequality (Cover & Thomas, 2012), marginalizing (or discarding some random variables) will not increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error can be obtained by combining Eq. (A.4) and Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality (=) would strictly hold only in the ideal case where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$, which means the diffusion model can accurately recover the underlying data distribution $q(\mathbf{x}_0)$. This upper bound for the overall KL divergence ($\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0))$) between true data and model-generated data is identical to the variational bound derived in DDPM (Ho et al., 2020) when data entropy $H(\mathbf{x}_0)$ is excluded, thereby suggesting that optimizing DDPM’s variational loss directly tightens this explicit upper bound on the discrepancy between trued data and model-generated data.

Diffused-prior gap. The first source of errors arises from the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ and the prior distribution used for sampling $p_\theta(\mathbf{x}_T)$. This mismatch stems from the fact that the number of diffusion steps T is finite in practice. The discrepancy can be formally quantified using a general distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence. Alternatively, other appropriate metrics such as FID or the Wasserstein distance can be employed to evaluate this distributional discrepancy. The diffused-prior gap, present from the outset, induces a deviation that persists throughout the denoising process, ultimately resulting in a sustained output error.

Noise model prediction error. The second source of errors stems from the noise prediction model. Because the model’s estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ , as guaranteed in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ represent this prediction error. Based on Eq. (5), any discrepancy in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\alpha_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Since each reverse step uses the mean from the previous iteration, this bias accumulates throughout the denoising trajectory, ultimately contributing to the overall prediction error.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Based on Eq. (7), we take a one-step numerical solver approximates the solution as example. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and starts from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Similarly, $\hat{\mathbf{x}}_k$ represent the numerical trajectory computed by a numerical ODE solver at discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$ and $\hat{h}$ is the estimated model function. For a specific numerical trajectory from $\hat{\mathbf{x}}_0 - \hat{\mathbf{x}}_N$, we want to calculate its total accumulated trajectory error. Let $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ be the trajectory error at step k . Consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Using the variation-of-constants formula for the exact solution over one step (from τ_k down to τ_{k+1}):

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. Note that the integration limits are backward in time.

Numerical path evolution. A one-step numerical solver approximates the solution. This numerical step deviates from the exact evolution governed by $\hat{h}$ starting from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ be the exact solution at τ_{k+1} of the ODE using $\hat{h}$, starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

So, we have:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtract the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12):

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. We can unroll this recurrence relation from $k = 0$ to $N - 1$. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Applying the recurrence

repeatedly:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial error term represents the deterministic impact on the final path error of a specific initial error instance, whose statistical origin lies in the distributional Diffused-Prior Gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As shown in Eq. (2), the forward process in DDPM (Ho et al., 2020) can be formulated as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

And the reverse process is defined as follows:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to that in Eq. (A.16), while its deterministic reverse process (with $\eta = 0$) is defined as follows:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share the same forward (diffusion) process, DDIM does not simply strip the stochastic term from DDPM's reverse (denoising) process. This is because the two methods employ different coefficient formulations for the predicted noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from the three datasets used in this study. As shown in Table A1, each dataset exhibits a balanced distribution across its respective categories. Table A2 further demonstrates that the distribution of images with different object counts is comparable, indicating the absence of object-count bias across datasets. The datasets appear in the same order in both tables, and the category order in Table A1 is defined as follows:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: Number of objects per category for each dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Number of images by object count(s) per dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We provide some representative examples of counting hallucinations corresponding to three datasets, as shown in Fig. A2. For the ToyShape and SimObject datasets, the counting hallucinations contain more than two instances of at least one category or do not contain any instances but with normal background, as shown in Fig. A2(a) and Fig. A2(b). In the Realhand dataset, counting-hallucinated samples are characterized by incorrect finger counts (e.g., 4 or 6 fingers), as shown in Fig. A2(c). Notably, the last two images of the first row in Fig. A2(c) appear to display five fingers; however, one finger is partially incomplete and visibly artifacted, effectively resulting in four valid fingers. Consequently, these samples are classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative examples of YOLO predictions on the RealHand dataset. The YOLO detector accurately identifies fingertips across diverse hand poses and orientations, including both dorsal and palmar views. However, it may still fail to recognize anatomically implausible configurations, such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but without a thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). These abcounting-valid samples are still detected as five-finger hands and therefore remain indistinguishable from anatomically valid ones in terms of finger counts.

Figure A3(b) shows several counting-hallucinated samples where YOLO predicts incorrect fingertip counts (e.g., 3, 4, 6, or 7 fingertips). Figure A3(c) further illustrates several rare edge cases (<1%) in which the detector fails to identify fingertips with unclear boundaries, cracked fingers, or floating artifacts. Such rare cases can be partly mitigated by tuning YOLO’s inference hyperparameters (e.g., IoU and confidence thresholds); however, doing so may compromise detection performance on the majority of counting-valid samples. Since a consistent detection setup is applied across all evaluations on the RealHand dataset, the presence of these rare cases does not affect the validity of our overall conclusions, even though addressing them remains a challenging and low-reward effort.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows representative examples of three classes of generated hand images: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, the evaluation procedure of the RealHand dataset first applies the counting-ready indicator to separate counting-ready samples (i.e., counting-valid and counting-hallucinated) from images unsuitable for counting. The latter typically exhibit severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers. Counting-ready samples are then further classified as counting-valid or counting-hallucinated based on whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we adopt 10 different hand-generation prompts to prompt Stable-Diffusion-3.5-Medium. The prompts are show as follows.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

The definition of hallucinations assumes that the model generates patterns that were never observed during training. However, this assumption is difficult to verify for large-scale pretrained diffusion models, where the full training distribution is inaccessible. To make the quantification of counting hallucinations in such cases feasible, we adopt a conservative criterion: we treat any generated hand image containing six or more fingers as a counting-hallucinated sample. This assumption is well justified, while real images may depict fewer than five visible fingers due to occlusion, articulation, or viewpoint, it is highly unlikely for real data to contain hands with six or more fingers. Thus, finger counts ≥ 6 provide a reliable and empirically verifiable signal of counting hallucination. Generated examples can be found in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

In the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, compared to 80k steps for the vanilla LDM. This increased training budget is necessary because extending the latent space from three to six channels in JDM introduces a substantial mismatch from the pretrained latent distribution of LDM. As a result, additional optimization is required to close this distributional gap and allow JDM’s modeling capacity to fully materialize.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Table A3: Comparison of failure rates and image quality between finetuning SD-1.5 and finetuning SD-1.5 with joint-diffusion training paradigm on RealHand. The first row (baseline) reports results from finetuning Stable-Diffusion-1.5 on RealHand data. The second row shows results obtained by freezing all intermediate layers of the baseline model and finetuning only the input and output layers with extended channel dimensions. Lower values indicate better performance.

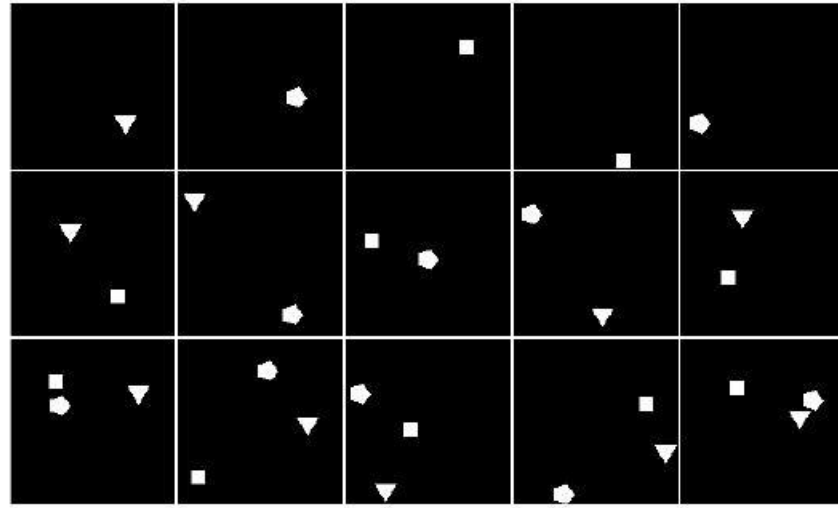
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
Sampling Steps																		
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

Importantly, JDM should be viewed as a **novel training paradigm** for improving factual generation in diffusion models rather than as a superior inference strategy. On the one hand, JDM only extends

the input and output channels of the UNet while keeping all sizes of intermediate layers unchanged, resulting in **introduces a negligible inference-time overhead compared to vanilla LDM**. On the other hand, extending a pretrained text-to-image model with additional mask channels **disrupts the manifold geometry of the pretrained latent space and underlines the original denoising distribution**, making direct finetuning such model nontrivial.

Despite this challenge, we still observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning conditions. In practice, we find that jointly aligning the extended latent space in JDM with the highly rigid latent geometry of large pretrained models is challenging under the limited scale of RealHand. To strike a practical balance, we adopt a two-stage finetuning strategy: we first finetune a Stable-Diffusion-1.5 model on RealHand, then freeze all intermediate UNet layers and finetune only the input and output layers with extended channels. As shown in Table A3, even under this highly constrained finetuning setup, the joint-diffusion paradigm consistently yields lower counting hallucination rates at 25 and 50 sampling steps across both solvers.

Overall, the evidence presented in this work suggests that JDM is a promising direction for enforcing factual consistency in image generation. Lastly, we believe that **exploring its behavior and performance under true large-scale training remains an important and compelling avenue for future research**.



(a) Examples in the ToyShape dataset

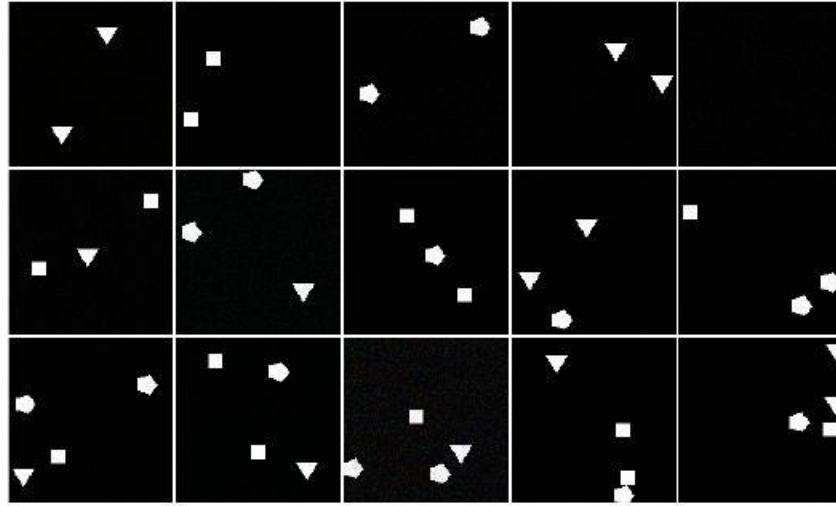


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

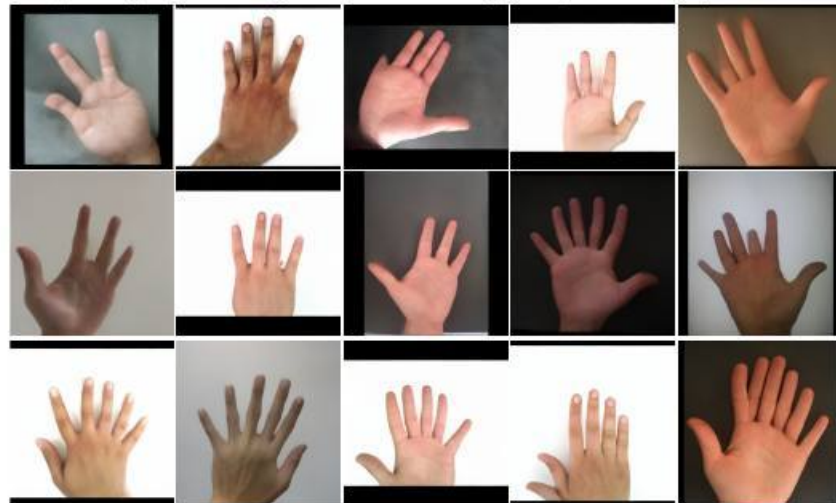
Figure A1: Examples of datasets in the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

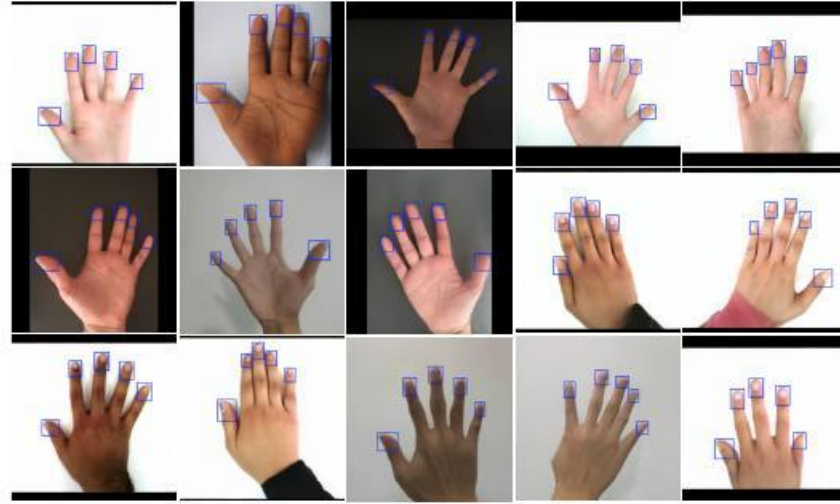


(b) Examples of counting hallucinations corresponding to the SimObject dataset

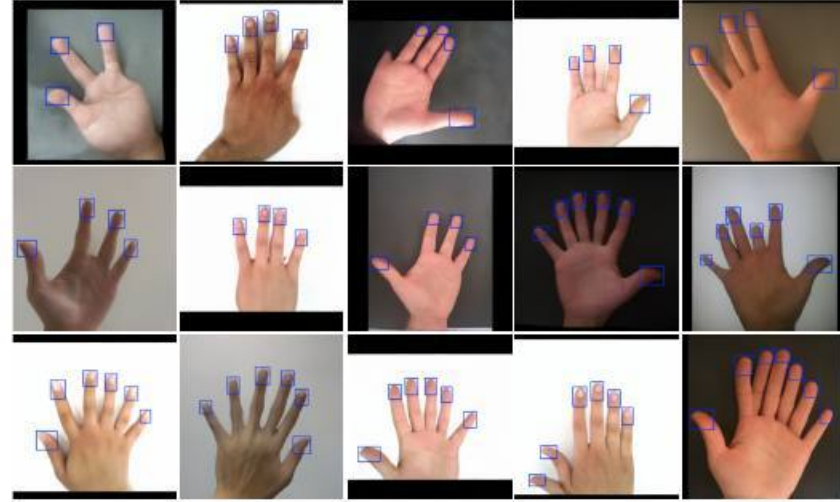


(c) Examples of counting hallucinations corresponding to the RealHand dataset

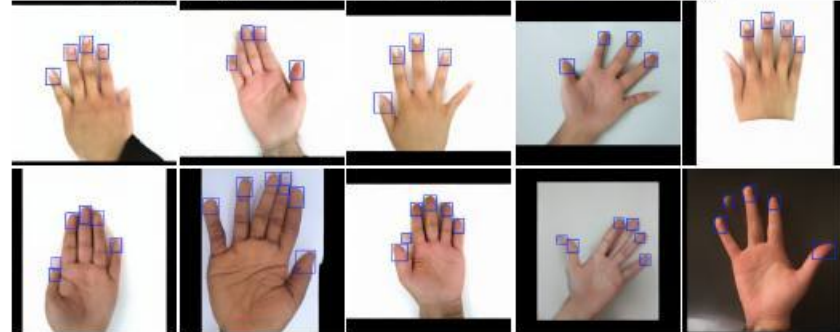
Figure A2: Examples of counting hallucinations corresponding to the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers

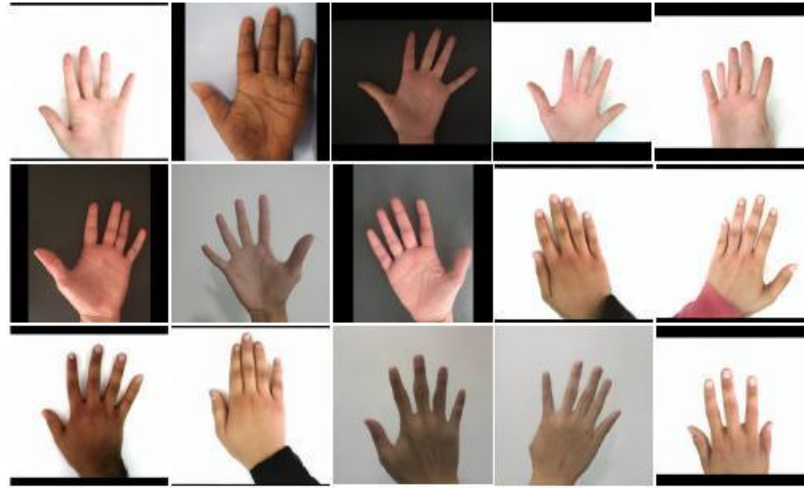


(b) Counting-hallucinated YOLO predictions with not five fingers

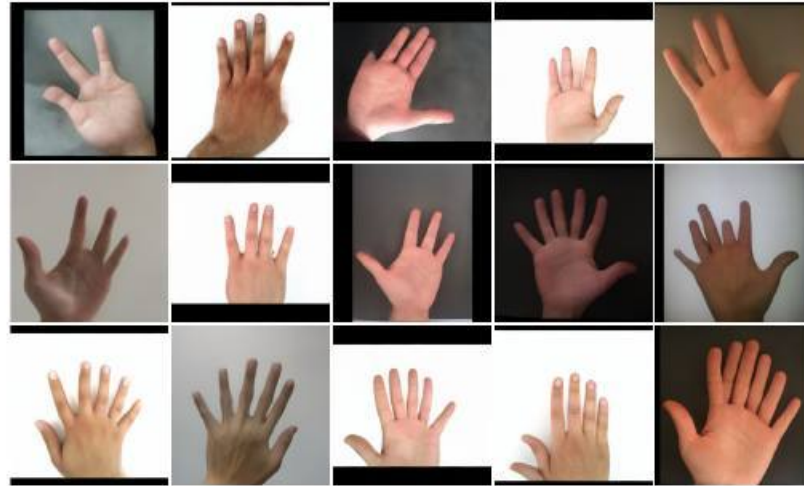


(c) Rare edge cases where YOLO predictions fail

Figure A3: Examples of YOLO predictions on generated RealHand images. (a) Counting-valid samples with five fingertip bounding boxes correctly predicted by YOLO (including cases with two thumbs). (b) Counting-hallucinated samples with incorrect fingertip count predictions (e.g., 3, 4, 6, or 7 detected fingertips). (c) Several rare edge cases where the counting-ready indicator fails to filter out and YOLO struggles to handle, such as blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: Examples of three classes of generated hand images. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: Examples of three classes of generated hand images using pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with more than five valid fingers (e.g., 6 or 7). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.